

Probabilistic Sports Prediction via Elo Ratings and Closing Line Value

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Gambling markets are among the most efficient information-aggregation mechanisms ever devised. Beating the closing line consistently is the proof of genuine predictive edge.
— Levitt (2004)

ChalkIQ is a multi-sport probabilistic prediction system generating calibrated win probabilities via Elo ratings, real-time injury signals, and live score diffusion. We evaluate across 1,672 games (Nov 2024 – Apr 2026), achieving a Brier score of 0.189 and log loss of 0.556 – improvements of 24% and 20% over the coin-flip baseline respectively.

The Elo Model

Each team carries a single rating $R \in \mathbb{R}$, initialised at 1500 and updated after every game. Win probability follows a logistic curve:

Definition: win probability

$$P(A \text{ wins}) = \frac{1}{1 + 10^{\frac{R_B - R_A - H}{\kappa}}}$$

where H is the home-court advantage and κ is the scale factor. A gap of κ points implies 75% win probability. For NCAAAB: $H = 100$, $\kappa = 300$.

After each game, the rating update incorporates both outcome and margin:

Definition: update rule

$$R'_A = R_A + K \cdot \text{MOV} \cdot (y - P(A \text{ wins}))$$
$$\text{MOV} = \frac{\ln(|m| + 1) \cdot 2.2}{\Delta_R \cdot 0.001 + 2.2}$$

K is the learning rate, m the point margin, Δ_R the pre-game rating gap. The logarithm dampens blowouts; the denominator corrects for autocorrelation – a dominant team routing a weak one gets a smaller update than evenly-matched teams separated by the same margin.

Remark. Parameters vary by sport. NCAAAB uses $K = 24$ (high roster turnover, volatile results). MLB uses $K = 6$ (even the best team loses 40% of games). Between seasons, ratings regress toward 1500 by a fraction ρ reflecting roster continuity: $\rho = 0.33$ for NCAAAB, 0.25 for NBA.

Closing Line Value

The primary evaluation metric is not win rate or P&L – it is Closing Line Value.

Definition: CLV

$$\text{CLV} = p_{\text{model}} - p_{\text{close}}$$

where p_{close} is the implied probability from the bookmaker's closing moneyline after removing the overround. Positive CLV means the model identified the correct side before the market fully priced it in.

Aside. CLV measures *prediction quality*, not outcome luck. The closing line aggregates sharp bettors, syndicate models, and late injury news. A model consistently beating it has demonstrated something real – analogous to measuring α via forward returns rather than a single trade's P&L.

Example: empirical CLV over 1,672 games

Metric	Value
Beat closing line	45.9%
Mean CLV	−0.78%

Metric	Value
Std deviation	20.0%
Range	−80% to +72%
Brier score	0.189 (baseline 0.250)
Log loss	0.556 (baseline 0.693)
Favourite accuracy	70.9%

Remark. Mean CLV is negative (−0.78%), as efficient market theory predicts: the closing line has absorbed injury reports, sharp money, and late lineup news that a pure Elo model cannot see. The wide standard deviation (20%) reveals where the value actually lives – not in systematic outperformance, but in large edges on specific games where Elo diverges sharply from the market.

Live In-Game Model

Pre-game probabilities are repriced during games using a score diffusion model (Stern 1994): basketball scoring is approximated as a random walk.

Definition: *live win probability*

With team A leading by d points and t minutes remaining:

$$P(A \text{ wins} \mid d, t) = \Phi\left(\frac{d}{\sigma\sqrt{t}} + \text{logit}(p_0) \cdot \frac{\sqrt{3}}{\pi}\right)$$

$\sigma \approx 2.0$ pts/ $\sqrt{\text{min}}$ for college basketball. At tip-off the Elo prior p_0 dominates; as $t \rightarrow 0$ the lead term drives $P \rightarrow 1$.

Backtest

Walk-forward backtest over the 2025–26 NCAA season – ratings built only from games completed *before* each prediction.

Example: 2025–26 NCAA backtest

Month	Bets	Win rate	ROI	Cumul P&L
Nov 2025	119	85%	+62%	+\$7,382
Dec 2025	187	86%	+64%	+\$19,418
Jan 2026	351	74%	+41%	+\$33,954
Feb 2026	269	65%	+25%	+\$40,654
Mar 2026	148	75%	+43%	+\$47,045

1,076 bets placed | 75.2% win rate | +43.5% ROI | Brier 0.174 | Log loss 0.523

Aside. Win rate declines across the season as the model converges – early games have high edge because Elo hasn’t yet separated the field. By February, ratings are tighter and the market harder to beat. The backtest is in-sample; live performance would be lower after accounting for hold percentages and line shopping.

Conclusion

A parsimonious one-number-per-team Elo model, calibrated on 1,136 games and evaluated across 1,672, produces well-calibrated multi-sport win probabilities. Brier score 0.189 and log loss 0.556 represent meaningful improvements over baseline. Mean CLV of −0.78% is consistent with market efficiency; the 20% standard deviation indicates genuine signal on individual games. The same framework – Elo ratings, CLV

measurement, score diffusion – ports across seven sports by changing only the parameters, suggesting the model captures real structure in competitive dynamics.

References

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